ASSIGNMENT 2: DRONE AERODYNAMICS, PROPULSION & ELECTRONICS

Full Name: Birendra Pratap Gautam

College Name: Rajkiya Engineering College Banda

Branch: Electrical Engineering

Year of Study: 3rd

Mobile Number: 8853746127

Email: birendrapratapgautam7@gmail.com

Internship Batch: B0826

Submission Date: 24/08/2026

Q1. Flight Stability Analysis

A quadcopter maintains a stable hover only when the forces and moments acting on it remain balanced. If it continuously drifts during flight, several aerodynamic or mechanical problems may be responsible.

1. Unequal Thrust Distribution

All four motors should produce properly balanced thrust during hover. If one motor produces less thrust because of wear, incorrect ESC calibration, or a damaged propeller, the drone tilts toward the weaker side. The tilted thrust vector develops a horizontal component, causing continuous sideways drift.

2. Incorrect Centre of Gravity

The centre of gravity should be close to the geometric centre of the drone. An incorrectly positioned battery, camera, or payload can shift the centre of gravity. This creates an imbalance, forcing the flight controller to continuously adjust motor speeds. A large offset can cause persistent drift and reduce stability.

3. Damaged or Mismatched Propellers

Bent, chipped, loose, or incorrectly matched propellers do not generate equal lift. They may also produce excessive vibration, which can affect sensor readings. As a result, the flight controller receives inaccurate motion information and the drone may drift, wobble, or struggle to maintain altitude.

4. Wind and Aerodynamic Disturbance

Crosswinds and turbulence exert horizontal forces on the drone. If the motors do not have sufficient reserve thrust to compensate, the drone moves in the direction of the wind. Turbulent airflow near buildings, walls, or other obstacles can make this behaviour more noticeable.

Conclusion

A stable hover requires balanced thrust, a correctly positioned centre of gravity, properly matched propellers, and sufficient thrust to overcome aerodynamic disturbances. Regular inspection and correct mechanical setup are therefore essential for safe and stable flight.

Q2. Electronics & Sensor Analysis

When GPS is lost during an autonomous mission, the drone depends heavily on its onboard flight controller and other sensors to remain stable.

Flight Controller

The flight controller acts as the main control unit of the drone. It receives data from sensors, processes the information, and continuously adjusts motor speeds through the ESCs. It is responsible for stabilization, navigation, and execution of flight commands.

IMU (Inertial Measurement Unit)

The IMU normally contains accelerometers and gyroscopes. The gyroscope measures rotational motion, while the accelerometer measures acceleration and helps estimate the drone's orientation. Together, they allow rapid correction of roll, pitch, and yaw.

GPS

GPS provides position, ground speed, and navigation information. It is important for autonomous missions, waypoint navigation, position hold, and return-to-home functions. If GPS is lost, accurate global position holding may become unavailable.

Barometer

The barometer measures atmospheric pressure and helps estimate altitude. It supports altitude-hold functions and allows the flight controller to maintain a more consistent flying height.

Magnetometer (Compass)

The magnetometer provides heading information relative to Earth's magnetic field. It helps the drone determine its direction and is particularly important during autonomous navigation.

Telemetry System

Telemetry provides a communication link between the drone and the ground station. It allows flight data, battery condition, position, and system warnings to be monitored during the mission.

Example of Two Component Failures

GPS Failure: The drone may lose accurate position hold and waypoint navigation. It may switch to a non-GPS stabilization mode and rely on the IMU, barometer, and pilot intervention.

IMU Failure: This is more critical because the flight controller may lose reliable attitude information. Incorrect roll, pitch, or yaw estimation can lead to severe instability and loss of control.

Conclusion

Stable flight depends on multiple sensors working together. Sensor fusion allows the flight controller to compare different measurements and maintain control even when one source, such as GPS, becomes temporarily unavailable.

BONUS CHALLENGE: Professional Flight Controller – Pixhawk 6C

Manufacturer

Holybro, based on the Pixhawk open hardware standard.

Key Features

Pixhawk 6C is a professional autopilot designed for reliable UAV control. It uses a powerful flight-management architecture, supports multiple communication interfaces, and can work with popular open-source autopilot software such as PX4 and ArduPilot.

Supported Sensors

It integrates inertial sensors such as accelerometers and gyroscopes and can interface with external GPS, magnetometers, barometers, airspeed sensors, rangefinders, and other navigation sensors.

Major Applications

It is used in multirotor drones, fixed-wing UAVs, VTOL aircraft, autonomous research platforms, mapping systems, inspection drones, and experimental robotic vehicles.

Conclusion

Its flexible hardware and broad sensor compatibility make it suitable for both educational projects and advanced autonomous UAV applications.